

TOTO Ltd.

TSE: 5332 — The Toilet Maker with a Semiconductor Engine

RECOMMENDATION: BUY

12-MO TARGET: ¥7,200 (+18%)

Reference price ¥6,097 (2 Oct 2026 close)	52-wk range ¥3,766–¥9,500	Mkt cap ~¥1.0tn (~US\$6.7bn)	Net cash (e) ~¥65bn
--	------------------------------	---------------------------------	------------------------

~51%

Ceramics share of FY3/26 segment OP — on 9% of sales

42.9%

Advanced Ceramics OP margin (electrostatic chucks for NAND cryo-etch)

¥7,200

12-mo target; probability-weighted value ~¥6,975

Buy: a ~43%-margin chip consumable priced as a toilet company, after a 36% pull-back

¥28.9bn Ceramics OP FY3/26 (+42%) — first year above the ¥670bn housing business	21.8x P/E FY3/27E; 17.3x FY3/28E — vs ¥9,500 peak (~27x)	~¥615bn Implied ceramics EV after valuing housing on peer regression (~15x FY3/28E EBIT)	¥7,200 12-mo SOTP target (+18%); P/E cross-check ¥7,216	25% Bear-case probability (memory capex downturn) — value ~¥3,500
--	--	--	---	---

Why now — four forces converging

- **Reset entry:** Q1 FY3/27 ESC shipment slip took the stock -11.4% on 3 Aug and -36% from the June high; FY guidance (OP ¥60bn) unchanged.
- **Capacity step-up:** Buzen kiln adds >20% ESC capacity in Jan 2027 into an accelerating NAND cryo-etch cycle (Lam: installed base to triple in 3–5 yrs).
- **Housing self-help:** China plants closed (~¥7bn improvement), largest Japan price rise in a decade (8–13%, 1 Dec 2026), Georgia plant ramping.
- **Activist round two:** Palliser won disclosure and capex; capital efficiency (~¥200bn excess capital) is the open item for STAGE3 (Apr 2027).

What has to go right

- H1 FY3/27 (~30 Oct) confirms the ceramics miss was timing — H1 ceramics OP ≥ ¥15bn.
- NAND WFE recovery continues (GS: \$11bn → \$15bn → \$22bn, 2026–28) and Lam keeps TOTO as the primary cryo-ESC source.
- Japan price rise sticks without share loss to LIXIL/Panasonic; China reaches break-even.
- STAGE3 plan sets ROE ≥10–12%, cross-holding exit and buyback framework.

Position sizing: event-driven, mid-weight. A ¥1tn Japanese large-cap that now behaves like a semiconductor-equipment supplier (±11% days on broker notes and results) — two very different investor bases own it, and the hand-off between them creates the volatility we want to exploit.

TOTO at a glance: two businesses, one P&L

What it does

- Japan's category-defining bathroom brand: ~60% of Japanese toilets; inventor of the Washlet (1980); >80% household bidet-seat penetration in Japan.
- ~72% of Japan housing sales are remodel/replacement — a steady but ~4%-margin annuity squeezed by costs and falling new-build.
- Since 1988 TOTO has made electrostatic chucks — the wafer-holding "table" inside etch tools; co-developed with Lam Research since ~1990; reported #2 globally.

Segment mix — FY3/26 (year to 31 Mar 2026)

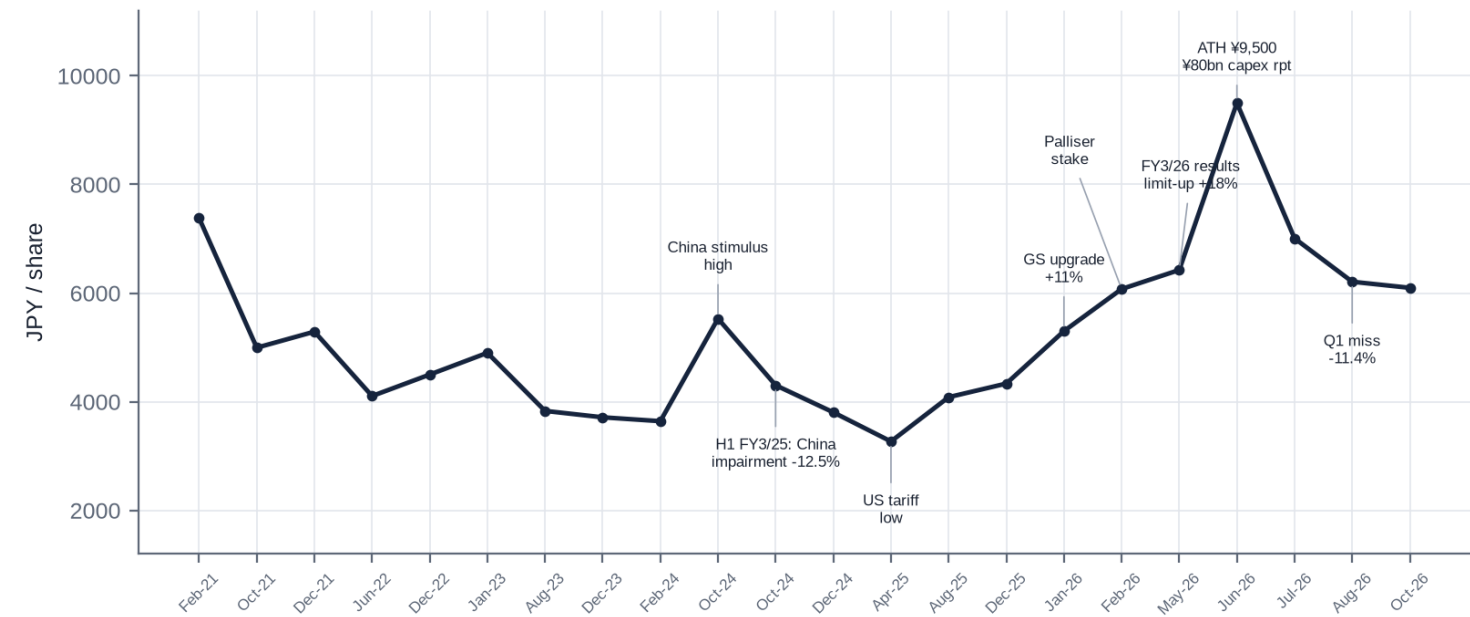
Segment	Sales	OP	Margin	Comment
Japan housing	¥479.7bn	¥20.3bn	4.2%	Remodel +1%, new-build -3%
China	¥53.8bn	-¥6.9bn	loss	2 plants closed 2025
Americas	¥75.6bn	~¥4.7bn	~6%	Georgia plant; tariffs
Asia/Oceania	¥54.9bn	¥10.2bn	18.6%	Vietnam, Taiwan, India
Advanced Ceramics	¥67.4bn	¥28.9bn	42.9%	+34% sales; ~51% of OP
Group	¥737.4bn	¥53.8bn	7.3%	Record OP

Where the power sits in the supply chain

- **Upstream:** clays, brass, resins, electronics; high-purity alumina/AlN powders — commodity inputs, cost inflation hits Japan margins.
- **Midstream:** vertically integrated kilns and ESC sintering (Kitakyushu, Oita/Nakatsu, Buzen, Chigasaki; Vietnam, Thailand, India, China, US, Mexico).
- **Downstream:** plumbing wholesalers → builders/remodelers (Japan); distributors (US/Asia); Lam Research for ESCs → Samsung, SK hynix, Kioxia, Micron.

The single most distinctive fact: 9% of revenue now produces over half the profit — and the two halves deserve multiples ~2x apart (housing ~9x EBIT, ceramics ~18x). The consolidated P&L hides this; the 2026 segment disclosure, won by the activist, exposes it.

A five-year round-trip: from China-beta to AI-memory momentum and back to earth



JPY/share; levels reconstructed from annual OHLC and reported event closes; illustrative between points.

Key catalysts

OCT 2024

China loss + ¥34bn impairment: -12.5% in a day.

APR 2025

US tariff low ¥3,269; ¥20bn buyback; China plant closures.

JAN-FEB 2026

Goldman upgrade (+11%); Palliser Capital Value Enhancement Plan.

1 MAY 2026

Record FY3/26, ceramics disclosure: limit-up +18%.

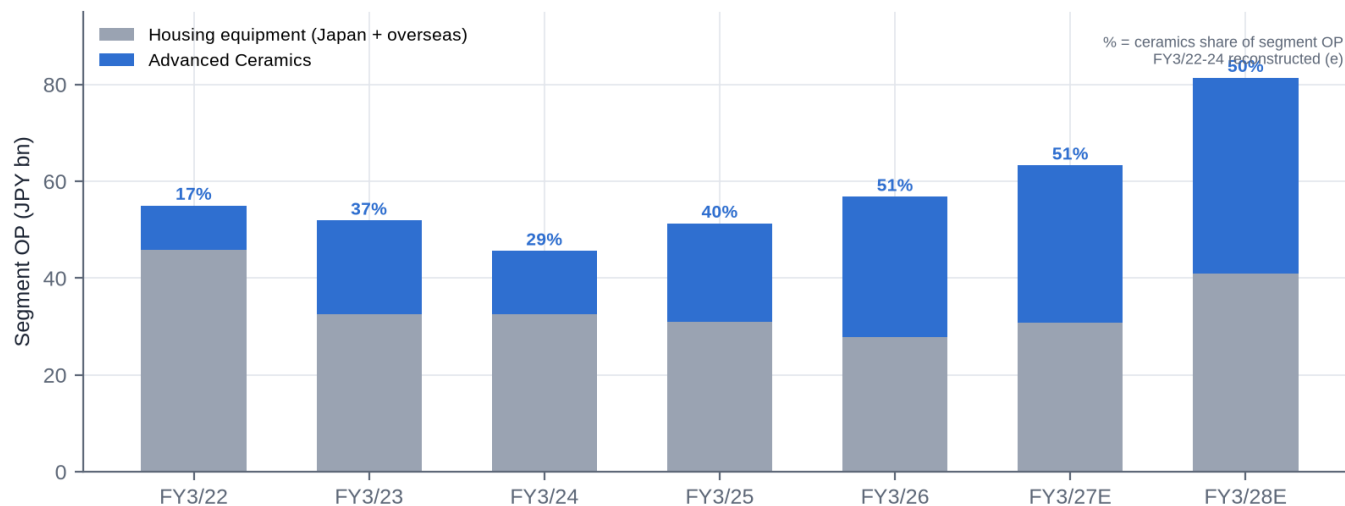
23 JUN 2026

ATH ¥9,500 on reported ¥80bn chip-parts programme.

3 AUG 2026

Q1 FY3/27 miss on ESC timing: -11.4%; now ¥6,097.

Ceramics went from 16% to ~51% of segment profit in four years



Why the ceramics franchise is real

- **Replacement annuity:** ~80% of segment sales are recurring chuck replacements (~1-year cycle, per Palliser).
- **Switching costs:** process-of-record part; requalification 12–24 months; 35-year Lam co-design loop; Lam Supplier Excellence 2023 and 2024.
- **Growth:** NAND layer counts → ~1,000 by 2030 makes cryo-etch indispensable; Buzen +20% (Jan 2027); ~¥30bn capex to FY2028, ~¥80bn 5-yr reported.
- **Margin:** 42.9% OP margin — 3x NGK Insulators, 2x Niterra.

But note FY3/24 (e): ceramics OP fell ~35% in the last memory downturn. This is a cyclical growth asset, not a bond.

Not on consolidated multiples — only once the parts are separated

Metric	TOTO	LIXIL	Rinnai	Takara	Masco	Geberit	NGK Ins.	Niterra	Kyocera
EV/Sales	1.27x	0.66x	0.75x	0.41x	2.1x	6.9x	~2.5x	~2.3x	~1.1x
EV/EBITDA	10.6x	9.0x	5.3x	n/a	11.6x	23.5x	12.0x	9.5x	7.8x
EV/EBIT	17.4x	~26x	~7x	5.4x	~12.5x	~29x	~17x	~12x	~20x
P/E (fwd)	21.8x	57x (t)	13.6x	13.8x	14.6x	29.8x	22.3x	~16x	16.7x
EBIT margin	7.3%	2.5%	10.7%	7.5%	16.8%	~24%	14.2%	18.9%	5.2%
Net debt/EBITDA	net cash	~4x	net cash	net cash	~1.9x	~1.0x	low	net cash	net cash

TOTO: EV/EBITDA, EV/EBIT on FY3/26A; P/E on our FY3/27E. Peers: latest FY / forward, Jul–Sep 2026, data aggregators; indicative (~ = derived).

What the comparison shows

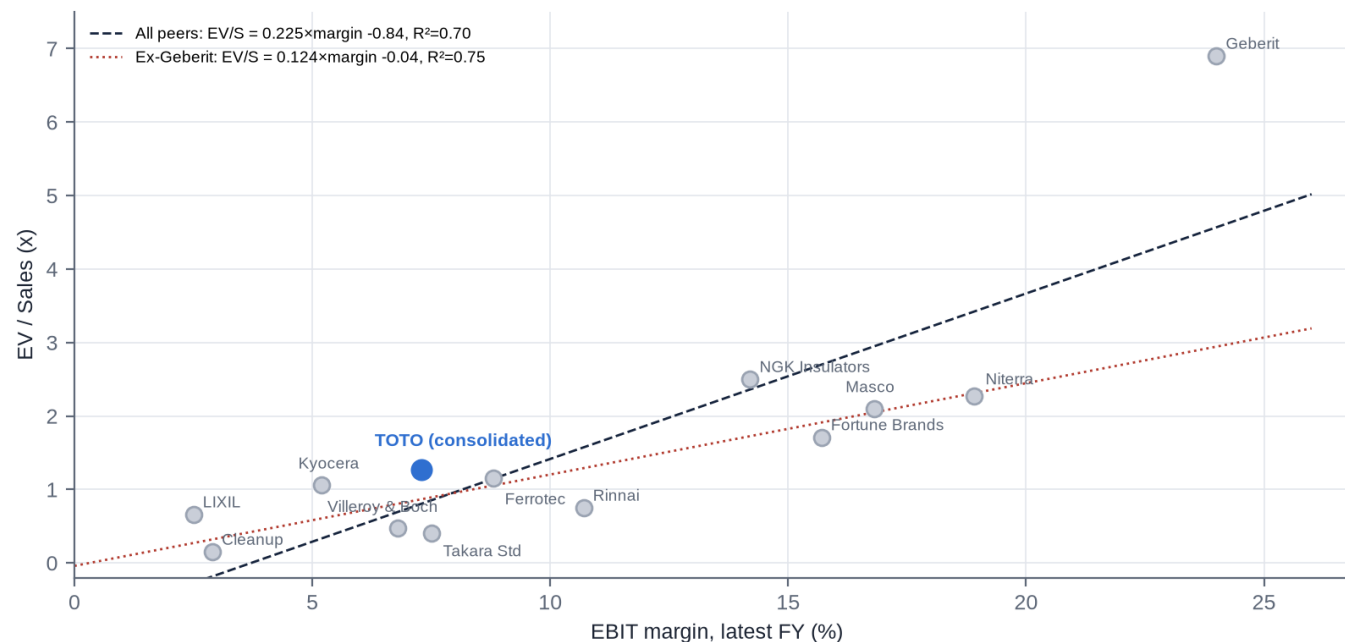
- On consolidated metrics TOTO trades at a premium to every Japanese housing peer (P/E 21.8x vs 13–14x) despite lower margins.
- It trades in line with technical-ceramics peers (NGK 22x fwd P/E) — yet ~65% of its sales are low-margin housing.
- Historical EV/EBITDA (10-yr median 13.7x) says cheap; P/B ~2.0x and P/E ~22x say fair.

The honest counter-read

- The "premium" is the market already paying for ceramics — it is not a mispricing of the whole.
- At ¥9,500 the stock discounted ~30x FY3/28E: the June peak was momentum, not fundamentals.
- Consensus EPS (~¥295) sits above company guidance (¥280) after a Q1 miss — near-term estimate risk remains.

The peer regression says expensive as a whole, fair-to-cheap as a sum of parts

EV/Sales vs EBIT margin — 12 listed peers



All peers: $EV/S = 0.225 \times \text{margin} - 0.84$, $R^2 = 0.70$; ex-Geberit: $EV/S = 0.124 \times \text{margin} - 0.04$, $R^2 = 0.75$. At TOTO's 7.3% consolidated margin the fitted EV/Sales is 0.81–0.87x vs 1.27x actual.

Reading the regression

- **Consolidated:** TOTO sits ~45–55% above the line — the statistical outlier is expensive, not cheap.
- **SOTP read:** housing (4.2% margin) on the line = $0.48 \times \text{sales} \approx \text{¥}320\text{bn EV}$. That leaves ~¥615bn for ceramics: ~9x sales, ~19x FY3/27E and ~15x FY3/28E EBIT.
- **Sanity check:** NGK trades ~17x EBIT on a 14% margin; a 43%-margin, 15–25%-growth ceramics consumable at ~15x FY3/28E is undemanding.
- **Implication:** the re-rating case depends on the market continuing to capitalise ceramics separately — which better disclosure (Palliser's ask) makes durable.

EPS +45% in two years: ceramics capacity plus housing self-help

JPY bn unless stated	FY3/26A	FY3/27E	FY3/28E	FY3/29E
Net sales	737.4	765	800	833
Ceramics sales	67.4	77.5	95.0	110.0
Ceramics OP (margin)	28.9 (42.9%)	32.5 (42%)	40.4 (42.5%)	46.2 (42%)
Housing OP (incl. China)	27.9	30.5	41.0	45.0
Group OP	53.8	60.0	78.0	87.5
OP margin	7.3%	7.8%	9.8%	10.5%
Net income	40.3	45.7	57.0	63.8
EPS (¥)	243	280	352	396
Consensus EPS (¥)	—	~295	~353	n/a
DPS (¥)	110	120	140	160

Reading the trajectory

- FY3/27 is a transition year: Q1 OP -1.6%, Middle East -¥3bn, Japan costs; we sit on guidance, below consensus.
- FY3/28 is the inflection: full-year Japan price rise, Buzen capacity, China at break-even.
- Sensitivity: ±10% ceramics sales = ±¥3.5bn OP = ±¥15 EPS.
- ROE ~7.7% → ~10% by FY3/28E — still below the ≥12% WILL2030 target without buybacks.

Our estimates; company FY3/27 guidance: sales ¥785bn, OP ¥60bn, NI ¥46bn, EPS ¥279.8, DPS ¥120 (unchanged at Q1).

A timing miss, punished like a thesis break

¥168.6bn Sales +1.7% vs ~¥176.5bn consensus (-4.5%)	¥8.07bn OP -1.6%; 13.5% of FY guide (normal ~18–20%)	¥42.0 EPS vs ¥45.6 consensus (-8%)	-11.4% Share-price reaction, 3 Aug 2026
---	--	--	---

What drove it

- Ceramics sales only +6% (OP +10%) vs +27% FY plan — chuck shipment timing.
- Japan housing OP -46%: procurement and labour costs, weak new-build.
- Middle East order suspension and costs: ~-¥3bn OP; FX gain +¥1.1bn below the line.
- Full-year guidance held; Dec 2026 price rise (8–13%) announced.

Sentiment tracker (management tone, 1–5)

Event	Focus	Tone
Oct 2024 (H1 FY3/25)	China loss, impairment	1.5
Apr 2025 (FY3/25)	Restructure, buyback	2.5
Oct 2025 (H1 FY3/26)	Ceramics surge	3.0
Apr 2026 (FY3/26)	Record, disclosure, capex	4.5
Jul 2026 (Q1 FY3/27)	Timing, costs, guide held	3.0

Signal: the stock was priced for uninterrupted acceleration. H1 results (~30 Oct) are the test of whether Q1 was timing (our base case) or the start of a NAND air-pocket.

Palliser won round one; capital efficiency is round two

17 Feb 2026 Palliser Capital publishes "Value Enhancement Plan" — top-20 holder (<5%)	>55% Upside claimed; intrinsic value ~¥8,800/share	30 Apr 2026 TOTO adopts the substance; Palliser "welcomes adoption" (7 May)	~¥200bn Returnable capital (excess equity, ~¥76bn cross-holdings, cash) (e)
---	---	---	---

Palliser demand	TOTO response	Status
Disclose Advanced Ceramics	Two-segment reporting; ceramics sales/OP/margin and FY plan	Achieved
Invest more in ceramics	~¥30bn capex to FY2028; Buzen +20%; ~¥80bn 5-yr reported	Achieved
Restructure low-profit housing / China	Plant closures (pre-dated campaign); ~¥7bn improvement targeted	Partial
Capital efficiency / equity ratio	DPS ¥110 → ¥120; buyback; cross-holdings <10% of net assets	Open

Assessment: no proxy fight was needed — a dispersed register (trust banks ~28%, life insurers ~9%) and TSE pressure made management receptive. The STAGE3 mid-term plan (~Apr 2027) is the natural venue for round two: ROE target, equity-ratio policy, cross-holding exit and a multi-year buyback. That is the special-situations catalyst.

A whole-company bid is unlikely; a ceramics structural option is not

Path	Probability (24m)	Indicative value	Precedent / logic
Whole-company take-private	5–10%	13–14x EBITDA → ¥8,000–8,600 (+30–40%)	FEFTA screening; stable holders; size ~US\$7bn
Ceramics IPO / minority sale / spin	10–15%	Ceramics ¥600–800bn (12–15x EBITDA)	Shinko → JIC (~¥685–700bn, ~12–13x EBITDA e)
Self-help: buybacks + cross-holding exit	50%+	+5–10% EPS accretion over 3 yrs	TSE P/B reform; Palliser
Status quo	remainder	SOTP ¥7,200	—

Relevant deal multiples (2021–26)

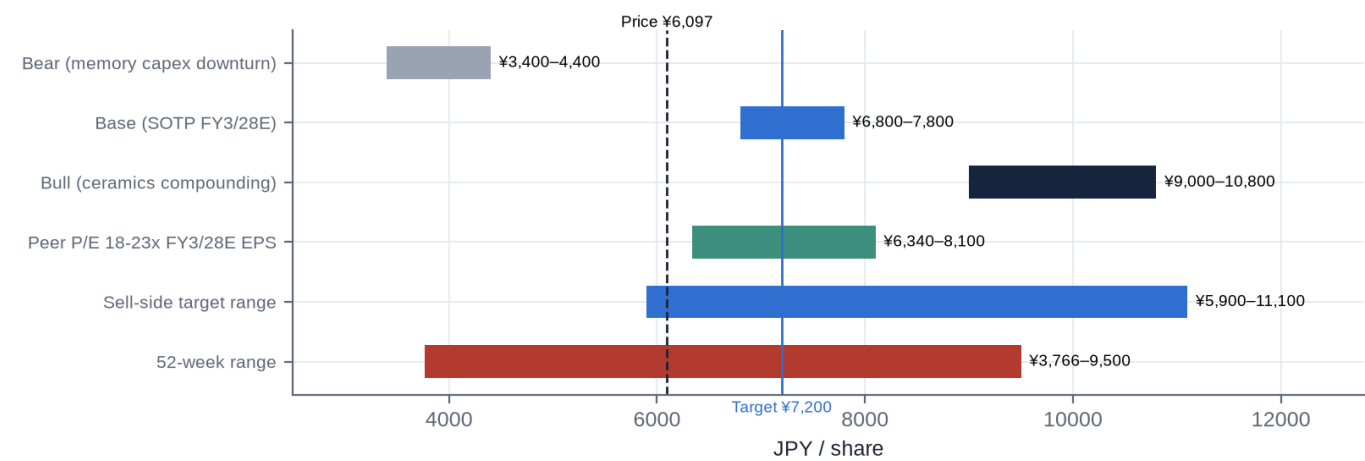
- Villeroy & Boch / Ideal Standard (2023): ~€600m, 8.1x EBITDA.
- Zurn / Elkay (2022): \$1.56bn, 14.2x EBITDA (9.8x post-synergy).
- RWC / EZ-Flo (2021): \$325m, 12x EBITDA.
- JIC / Shinko Electric (2023–25): ~¥685–700bn incl. Fujitsu stake.

Impediments

- FEFTA: semiconductor parts are a "core" sector — foreign bidders face review.
- Lam change-of-control risk on the ESC franchise.
- Morimura-group heritage, lifetime-employee management, no M&A track record (organic grower since 2016).

Asymmetry is acceptable, not compelling: +64% / -43%

Scenario	Prob.	Key assumptions	EPS FY3/29E	Multiple	Value
Bull	25%	Ceramics ¥125bn sales (+23% CAGR), 44% margin; housing OP ¥50bn; buybacks	~¥470	Ceramics 22x EBIT; housing 12x	~¥10,000 (+64%)
Base	50%	Ceramics ¥110bn (+18% CAGR), 42%; China break-even; price rise sticks	~¥396	SOTP 18x / 9x (FY3/28E)	~¥7,200 (+18%)
Bear	25%	NAND capex downturn: ceramics flat ~¥60bn at 35%; housing ¥26bn	~¥220	Ceramics 12x; housing 8x; ~16x P/E	~¥3,500 (-43%)
Prob.-weighted					~¥6,975 (+14%)



What moves us between cases

- Lam/NAND WFE commentary (21 Oct) and TOTO H1 ceramics (~30 Oct).
- Japan volumes after the 1 Dec price rise.
- STAGE3 capital-return framework (Apr 2027).
- Any evidence of Lam dual-sourcing or TEL cryo-etch wins.

One technology, one customer, one end market — and a 4%-margin anchor

Dismantling the bull case

- **Concentration:** >50% of group OP from cryo-ESCs, largely sold to Lam Research for NAND.
- **Most underestimated competitor:** Tokyo Electron — its cryo-etch success would reallocate the installed base away from the Lam platform TOTO is embedded in.
- **Moat weaker than it looks:** AMAT/Lam in-house ESCs; JIC-backed Shinko; NGK/Kyocera scale in AIN. 43% margins invite price pressure.
- **Worst capital allocation:** China — ~¥49bn written off 2024–26; persistent over-capitalisation (equity ratio ~64%, ROE <8%).
- **Growth -20–30%:** EPS ~¥300–320 by FY3/29 → ¥5,100–5,400 at 17x; ~¥4,300 at 14x.

Accounting / forensic flags

Area	Flag	Risk
Revenue recognition	Shipment-based; Q1 "timing" lumpiness	Med
Segment reporting	3 → 2 segments obscures China; cost allocation opaque	Med
Impairments	¥34bn + ~¥15bn China "one-offs"	Med
Earnings quality	Ordinary > OP by ¥6.9bn (FX, dividends)	Low–Med
Provisions	Washlet/faucet recalls	Low–Med
Incentives	OP-linked cash bonus; little equity pay	Med
Goodwill / SBC / related parties	Minimal / none identified	Low

Single permanent-impairment scenario: NAND moves to a non-cryogenic high-aspect-ratio etch route (or TEL wins the next node) while Lam qualifies a second ESC source. We put this at ~10–15% over three years.

Under-levered, over-capitalised — the self-help lever

63.8% Equity ratio (Mar 2026) — reported	~¥65bn Net cash (e); net debt/EBITDA ≈ -0.7x	¥60–76bn Cross-shareholdings (e); target <5% of net assets by FY2030	¥120 FY3/27 DPS guide (payout ≥40%, progressive); ¥20bn buyback 2025 cancelled
--	--	---	--

Top holders (approx., Mar 2026)

Holder	Type	%
Master Trust Bank of Japan	Custodian	~17–19%
Custody Bank of Japan	Custodian	~9.8%
Meiji Yasuda Life	Life insurer	~6.1%
Nippon Life	Life insurer	~3.2%
State Street (2 a/c)	Foreign custodian	~3.9%
Employee Stock Ownership	Insider	~2.0%
Nomura AM group (5% rpt)	Institutional	~8.0%
Palliser Capital	Activist	<5%

Management & board

- President Shinya Tamura (since Apr 2025; ex-US/Vietnam head); Chairman Noriaki Kiyota (President 2020–25); CFO Tomoyuki Taguchi.
- Professional lifetime managers; negligible insider ownership; pay ~¥140–170m for top executives, OP-linked.
- Board cut 13 → 10 at June 2026 AGM, 50% independent.
- Short interest low (~1–2% e); reporters mostly quant/market-makers (Millennium, Qube, UBS, MS MUFG).

Lam, H1 results and the price rise dominate the next two quarters

21 OCT 2026	Lam Research Sep-quarter results NAND/cryo-etch demand read-across for ESC orders.	HIGH
~30 OCT 2026	TOTO H1 FY3/27 results Was Q1 timing? Ceramics H1 OP, FY guidance, consensus reset.	CRITICAL
1 DEC 2026	Japan price increase (8–13%) Pre-buying in Oct–Nov; elasticity and competitor follow-through.	HIGH
JAN 2027	Buzen ESC kiln complete; Q3 results (~29 Jan) +20% capacity online; seasonal Q3 profit test.	HIGH
~LATE APR 2027	FY3/27 results + STAGE3 mid-term plan ROE, equity ratio, buybacks, cross-holding exit — activist round two.	CRITICAL
ONGOING	US tariffs, Palliser / 5% filings, cross-holding sales Americas margin; activist re-engagement.	HIGH

What would make this thesis wrong

01 Memory capex downturn NAND makers pause after the 2026 upgrade wave; ceramics OP falls ~35% as in FY3/24 and the multiple compresses at the same time.	02 Customer concentration Lam qualifies a second ESC source or loses cryo-etch share to Tokyo Electron — the franchise economics reset permanently.	03 Japan housing erosion Starts fall toward ~600k; the Dec 2026 price rise loses share; costs keep Japan margins near 3–4%.
04 China relapse Break-even slips again; further impairments on remaining Zhangzhou/Dalian assets.	05 Capital hoarding STAGE3 prioritises the ¥1tn sales target and capex over ROE and returns; activist disengages.	06 Tariffs and FX US tariffs on Vietnam/Mexico/Thai/Malaysian sourcing; yen strength cuts translation and ceramics export margins.

Engage on capital allocation and ceramics transparency, alongside Palliser

Top questions for the CEO / Chairman

- What share of ceramics revenue is Lam, and are you sole- or dual-source on its cryo platforms?
- Was the Q1 shortfall purely timing — what are H1 ceramics orders and book-to-bill?
- Are you qualified with Tokyo Electron's cryogenic etch?
- Will STAGE3 include ROE $\geq 10\text{--}12\%$, equity-ratio and TSR targets?
- How much of the cross-shareholding book will be sold by FY3/29?
- Would the board consider structural options for ceramics if the SOTP gap persists?

Sizing & monitoring

- Mid-weight, event-driven: build into H1 results (~30 Oct); add on confirmation of ceramics re-acceleration.
- Hard trigger: reassess if H1 ceramics OP is down y/y with no timing explanation, or FY ceramics plan (¥85.5bn) is cut.
- Track Lam (21 Oct) and AMAT (~mid-Nov) commentary on NAND cryo-etch as leading indicators.
- Pair option: long TOTO / short Japanese housing basket (LIXIL, Takara) isolates the ceramics re-rating.

Recommendation: BUY

¥7,200

12-mo target (+18%); prob.-weighted ~¥6,975

~¥615bn

Implied ceramics EV — ~15x FY3/28E EBIT

21.8x

P/E FY3/27E; 17.3x FY3/28E

~¥200bn

Returnable capital — activist round two

- TOTO's consolidated multiples look full, but the peer regression shows the premium is the market paying for ceramics — housing alone is worth only ~¥320bn.
- At ¥6,097 the ~43%-margin ESC franchise is capitalised at ~15x FY3/28E EBIT, after a timing-driven 36% pull-back from the June high.
- Housing self-help (China break-even, Dec 2026 price rise, US ramp) adds non-cyclical earnings the momentum sell-off ignored.
- The activist has won disclosure and capex; capital efficiency in the April 2027 STAGE3 plan is the next special-situations catalyst.

Sizing & monitoring

Mid-weight, event-driven: a large-cap with semiconductor-style volatility and a 25% bear case (memory downturn, ~¥3,500).

Hard trigger: reassess if H1 FY3/27 (~30 Oct) shows ceramics OP down y/y without a timing explanation.

Catalyst watch: Lam (21 Oct), H1 results, 1 Dec price rise, Buzen capacity (Jan 2027), STAGE3 plan (Apr 2027).